

Lease Management

College: 4216-Mailam engineering college

Team ID:NM2024TMID00673

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1. Project Overview

This Lease Management project is designed to optimize and automate the processes involved in managing lease agreements for real estate properties, equipment, or other assets. By leveraging Salesforce's platform, the project aims to streamline lease management tasks, improve communication, and ensure compliance with regulations. The system will facilitate efficient tracking, communication, and record-keeping for all lease-related transactions, enhancing both operational efficiency and data accuracy.

2.Objectives

Business Goals:

- To create a streamlined and automated system for lease management that supports efficient tracking, compliance, and tenant relations.

- The project aims to reduce administrative overhead, improve communication with tenants, and ensure compliance with lease terms and regulations.

Specific Outcomes:

- A comprehensive dashboard to track lease terms, payments, and tenant details.
- Automated reminders and notifications for lease renewals, payment due dates, and tenant communications.
- Efficient approval processes and validation checks to ensure data accuracy and compliance.

3. Salesforce Key Features and Concepts Utilized

This project uses key Salesforce functionalities to develop a robust lease management system:

1. Custom Objects and Relationships
 - Created custom objects such as Property, Tenant, Lease, and Payment to manage essential lease data.
 - Established lookup relationships among objects to connect lease records with tenants and properties, supporting easy data retrieval and record linkage.
2. Tabs and Lightning App (Lease Management)
 - Created tabs for each custom object, making navigation easier within the Lease Management Lightning app.
 - Customized the app to organize functionalities for streamlined access and branding.
3. Validation Rules
 - Developed validation rules to maintain data integrity, ensuring accurate entries in the Lease object and preventing incomplete or incorrect submissions.
4. Email Templates
 - Created custom email templates to support tenant communications:
 - Tenant Leaving Notification
 - Leave Approved
 - Leave Rejection
 - Monthly Payment Reminder
 - Successful Payment Confirmation
5. Approval Process
 - Configured an approval process to check for vacant properties:
 - Initial Submission Action: Sends for review upon submission.

- Final Approval Action: Marks the property as leased and initiates tenant communication.
 - Final Rejection Action: Triggers notification to tenant with rejection details.
6. Apex Triggers and Classes
 - Developed an Apex Trigger to automate key lease management tasks.
 - Created an Apex handler class to manage specific functions within the system, including the validation of lease terms and calculation of payment due dates.
 7. Flows
 - Designed a Monthly Payment Flow to streamline payment processing for tenants, supporting reminders and status tracking.
 8. Scheduled Apex Class
 - Created an Apex class for scheduled operations, automating monthly billing reminders and payment checks to ensure timely notifications and compliance.

4. Detailed Steps to Solution Design

1. Created Salesforce Developer Account

- We began by creating a Salesforce Developer Account to access a development environment where we could build and customize the application. This setup provided the necessary tools to design, test, and deploy the application.

Build enterprise-quality apps fast to bring your ideas to life

- Build apps fast with drag and drop tools
- Customize your data model with clicks
- Go further with Apex code
- Integrate with anything using powerful APIs
- Stay protected with enterprise-grade security
- Customize UI with clicks or any leading-edge web framework

Sign up for your Salesforce Developer Edition

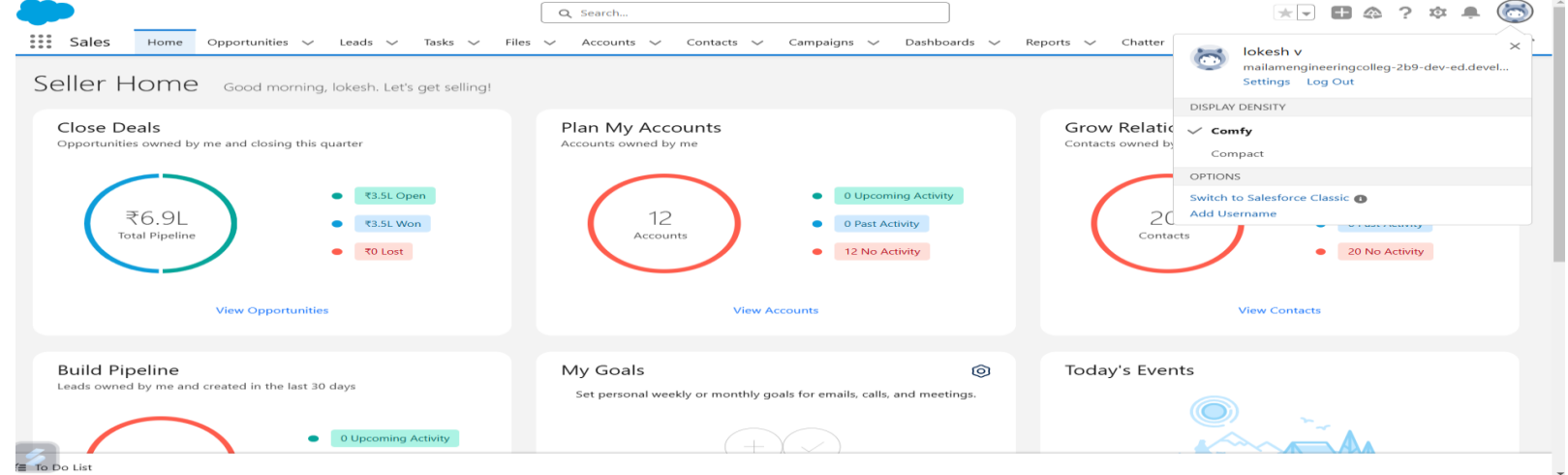
A Salesforce Platform environment for free.

Complete the form to get access to the Salesforce Developer Edition.

First Name*	Last Name*
nithishkumar	B
Email*	
nithishkumar3989@gmail.com	
Role*	
Developer	
Company*	
mallam engineering college	
Country/Region*	
India	
State/Province*	
Tamil Nadu	
Postal Code*	
603401	
Username*	
leasemanagment@gmail.com	

Your username must be in the form of an email address (it does not have to be real). It must be unique and cannot be associated with another Salesforce login credential. [Read more about username recommendations.](#)

☐ I agree to the [Main Services Agreement](#) – Developer Services and



2. Object Definitions

- Defined primary objects: **Property**, **Tenant**, **Lease**, and **Payment** within Object Manager.
- Each object was designed to store specific data: Property details, tenant information, lease agreements, and payment records.

Custom Object Information

The singular and plural labels are used in tabs, page layouts, and reports. Be careful when changing the name or label as it may affect existing integrations and merge templates.

Label: Example: Account
 Plural Label: Example: Accounts
☐ Starts with vowel sound

The Object Name is used when referencing the object via the API.
 Object Name: Example: Account

Description:

Context Sensitive Help Setting: ☒ Open the standard Salesforce.com Help & Training window
☐ Open a window using a Visualforce page
 Content Name:

Enter Record Name Label and Format

The Record Name appears in page layouts, key lists, related lists, lookups, and search results. For example, the Record Name for Account is "Account Name" and for Case it is "Case Number". Note that the Record Name field is always called "Name" when referenced via the API.

Record Name: Example: Account Name
 Data Type: Warning: if you plan to insert a high volume of records in this object, via the API for example, use the Text data type.

Optional Features

☒ Allow Reports

Custom Object Information

The singular and plural labels are used in tabs, page layouts, and reports.

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Enter Record Name Label and Format

The Record Name appears in page layouts, key lists, related lists, lookups, and search results. For example, the Record Name for Account is "Account Name" and for Case it is "Case Number". Note that the Record Name field is always called "Name" when referenced via the API.

Record Name: Example: Account Name
 Data Type: Warning: if you plan to insert a high volume of records in this object, via the API for example, use the Text data type.

Optional Features

☒ Allow Reports
☒ Allow Activities

3. Tab Configuration

- Configured tabs for each main object to facilitate quick access.
- Enabled tabs for Property, Tenant, Lease, and Payment, allowing users to manage lease data seamlessly.

Custom Tabs

You can create new custom tabs to extend Salesforce functionality or to build new application functionality. Custom Object tabs look and behave like the standard tabs provided with Salesforce. Web tabs allow you to embed external web applications and content within the Salesforce window. Visualforce tabs allow you to embed Visualforce pages. Lightning Component tabs allow you to add Lightning components to the navigation menu in Lightning Experience and the mobile app. Lightning Page tabs allow you to add Lightning Pages to Lightning Experience and the mobile app.

Action	Label	Tab Style	Description
EDIT DEL	Encasement Services	Guided Tour	
EDIT DEL	LEASE	Blank	
EDIT DEL	Payment	Blank	
EDIT DEL	property	Blank	
EDIT DEL	Research Proposal	Blank	
EDIT DEL	Tenants	Blank	

Web Tabs

No Web Tabs have been defined.

Visualforce Tabs

No Visualforce Tabs have been defined.

4. Lightning App Development

- Created a custom Lightning App named **Lease Management** to consolidate all functionalities.
- Configured branding, navigation items, and access settings to improve user experience and accessibility.

New Lightning App

App Details & Branding

Give your Lightning app a name and description. Upload an image and choose the highlight color for its navigation bar.

App Details

*App Name: Lease Management

*Developer Name: Lease_Management

Description: Enter a description...

App Branding

Image: [Upload button]

Primary Color Hex Value: #0070D2

Org Theme Options: ☐ Use the app's image and color instead of the org's custom theme

App Launcher Preview

Next

Lightning Experience App Manager

22 items • Sorted by App Name • Filtered by All app menu items - TabSet Type: App Type

App Name	Developer Name	Description	Last Modified Date	App Type	Visible
1 All Tabs	AllTabSet		19/09/2024, 10:21 pm	Classic	✓
2 Analytics Studio	Insights	Build CRM Analytics dashboards and apps	19/09/2024, 10:21 pm	Classic	✓
3 App Launcher	AppLauncher	App Launcher tabs	19/09/2024, 10:21 pm	Classic	✓
4 Bolt Solutions	LightningBolt	Discover and manage business solutions designed for your industry.	19/09/2024, 10:21 pm	Lightning	✓
5 Community	Community	Salesforce CRM Communities	19/09/2024, 10:21 pm	Classic	✓
6 Content	Content	Salesforce CRM Content	19/09/2024, 10:21 pm	Classic	✓
7 Data Manager	DataManager	Use Data Manager to view limits, monitor usage, and manage recipes.	19/09/2024, 10:21 pm	Lightning	✓
8 Digital Experiences	SalesforceCMS	Manage content and media for all of your sites.	19/09/2024, 10:21 pm	Lightning	✓
9 Hive App	Hive_App		20/09/2024, 7:04 pm	Lightning	✓
10 Lease Management	Lease_Management		07/10/2024, 6:38 pm	Lightning	✓
11 Lightning Usage App	LightningInstrumentation	View Adoption and Usage Metrics for Lightning Experience	19/09/2024, 10:21 pm	Lightning	✓
12 Marketing CRM Classic	Marketing	Track sales and marketing efforts with CRM objects.	19/09/2024, 10:21 pm	Classic	✓

5. Custom Field Creation

- Created custom fields within each object to capture required details:
 - **Property:** Location, type, rental amount, availability status.

property | Salesforce

psg-f-dev-ed.develop.lightning.force.com/lightning/setup/ObjectManager/01dM0000028E6P/FieldsAndRelationships/v...

Setup Home Object Manager

property

Details

Fields & Relationships

8 Items, Sorted by Field Label

Quick Find New Deleted Fields Field Dependencies Set History Tracking

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Address	Address__c	Long Text Area(32768)		
Created By	CreatedById	Lookup(User)		
Last Modified By	LastModifiedById	Lookup(User)		
Name	Name__c	Text(25)		
Owner	OwnerId	Lookup(User,Group)		✓
property Name	Name	Text(80)		✓
slqt	slqt__c	Text(18)		
Type	Type__c	Picklist		

Page Layouts
Lightning Record Pages
Buttons, Links, and Actions
Compact Layouts
Field Sets
Object Limits
Record Types
Related Lookup Filters
Search Layouts
List View Button Layout
Restriction Rules
Scoping Rules
Triggers

- **Tenant:** Contact information, lease start/end dates.

Tenant | Salesforce

psg-f-dev-ed.develop.lightning.force.com/lightning/setup/ObjectManager/01dM0000028E81/FieldsAndRelationships/v...

Setup Home Object Manager

Tenant

Details

Fields & Relationships

7 Items, Sorted by Field Label

Quick Find New Deleted Fields Field Dependencies Set History Tracking

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Created By	CreatedById	Lookup(User)		
Email	Email__c	Email		
Last Modified By	LastModifiedById	Lookup(User)		
Owner	OwnerId	Lookup(User,Group)		✓
Phone	Phone__c	Phone		
status	status__c	Picklist		
Tenant Name	Name	Text(80)		✓

Page Layouts
Lightning Record Pages
Buttons, Links, and Actions
Compact Layouts
Field Sets
Object Limits
Record Types
Related Lookup Filters
Search Layouts
List View Button Layout
Restriction Rules
Scoping Rules
Triggers

- **Lease:** Lease terms, conditions, and expiration date.

SETUP > OBJECT MANAGER

lease

Details

Fields & Relationships

6 Items, Sorted by Field Label

Quick Find

New Deleted Fields Field Dependencies Set History Tracking

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Created By	CreatedById	Lookup(User)		
End date	End_date__c	Date		
Last Modified By	LastModifiedById	Lookup(User)		
Lease Name	Name	Text(80)		✓
Owner	OwnerId	Lookup(User,Group)		✓
start date	start_date__c	Date		

- **Payment:** Amount due, due date, and payment status.

SETUP > OBJECT MANAGER

Payment for tenant

Details

Fields & Relationships

7 Items, Sorted by Field Label

Quick Find

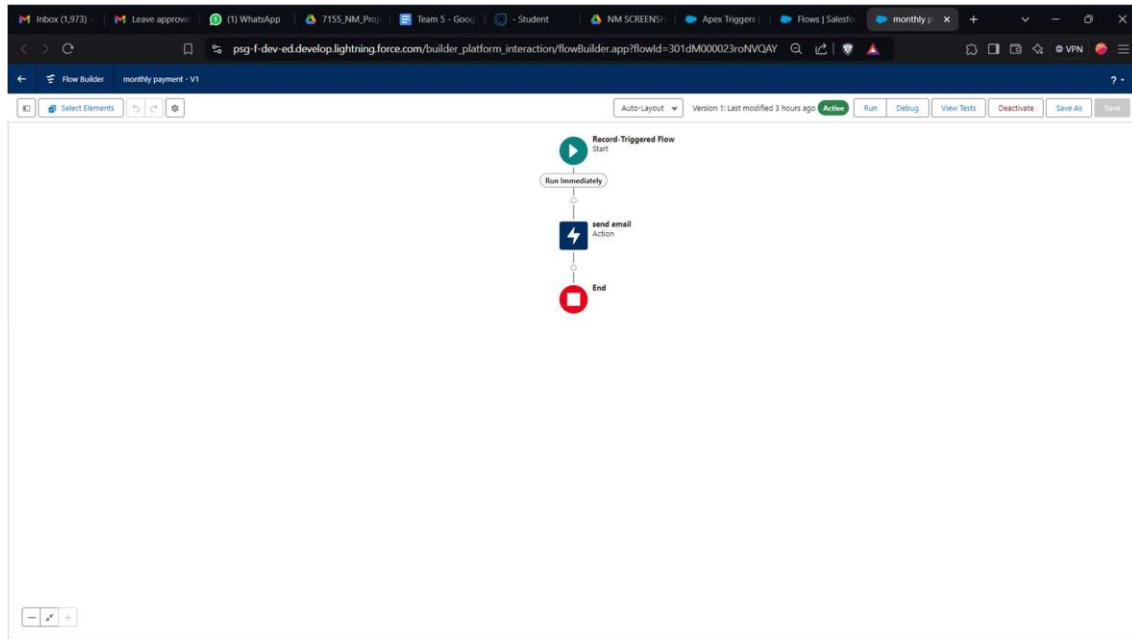
New Deleted Fields Field Dependencies Set History Tracking

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Amount	Amount__c	Number(18, 0)		
check for payment	check_for_payment__c	Picklist		
Created By	CreatedById	Lookup(User)		
Last Modified By	LastModifiedById	Lookup(User)		
Owner	OwnerId	Lookup(User,Group)		✓
Payment date	Payment_date__c	Date		
Payment Name	Name	Text(80)		✓

6. Flow Creation for Monthly Payments

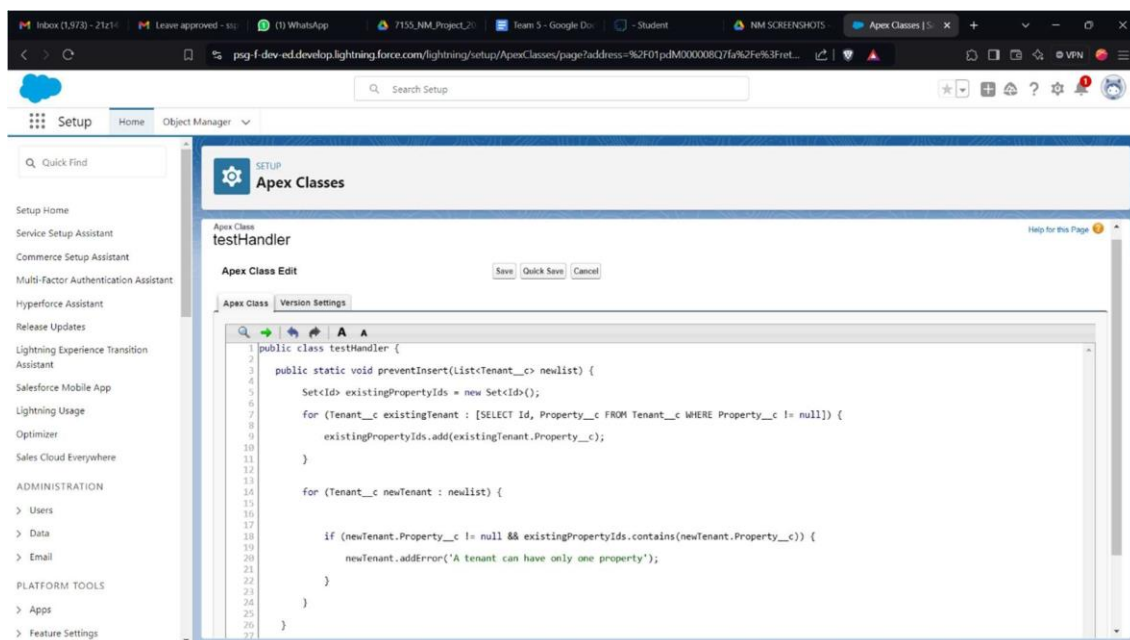
- Developed a **Monthly Payment Flow** to guide users through the payment process for tenants.

- Included automated reminders and status tracking, streamlining monthly rent collection.



7. Apex Trigger and Handler Class

- Created an Apex Trigger to automate lease calculations, such as due dates and reminders.
- Developed an Apex handler class to process complex calculations and manage lease status changes.



8. Email Template Setup

- Designed email templates for tenant communication, including notifications for payment reminders, successful payments, and lease approval or rejection.

The screenshot shows the Salesforce Classic Email Template Setup interface. The left sidebar contains navigation links: Setup, Home, Object Manager, and a search bar. The main content area is titled "Classic Email Templates" and shows a "Text Email Template" named "tenant leaving". Below the title, there is a preview of the email template. The "Email Template Detail" section shows the following information:

Email Templates from Salesforce	Unfiled Public Classic Email Templates
Email Template Name	tenant leaving
Template Unique Name	tenant_leaving
Encoding	Unicode (UTF-8)
Author	lokesh v [Change]
Description	
Created By	lokesh v, 16/11/2024, 11:58 am
Modified By	lokesh v, 16/11/2024, 11:58 am

The "Email Template" section shows the subject "request for approval the leave" and the plain text preview:

Dear({Tenant__c.CreatedBy}).

Please approve my leave

The screenshot shows the Salesforce Classic Email Template Setup interface for a "Text Email Template" named "Leave approved". The left sidebar contains navigation links: Setup, Home, Object Manager, and a search bar. The main content area is titled "Classic Email Templates" and shows a "Text Email Template" named "Leave approved". Below the title, there is a preview of the email template. The "Email Template Detail" section shows the following information:

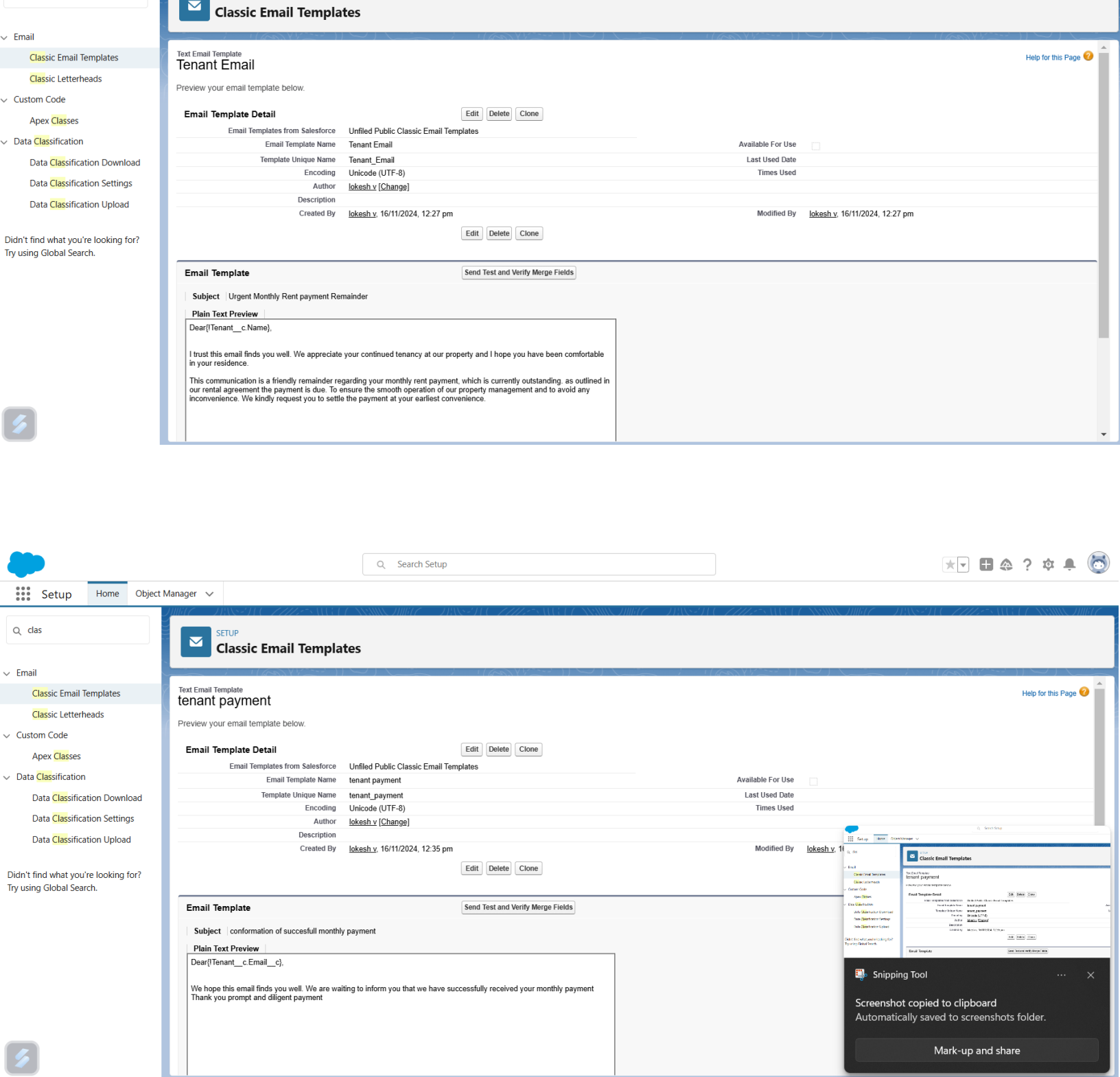
Email Templates from Salesforce	Unfiled Public Classic Email Templates
Email Template Name	Leave approved
Template Unique Name	Leave_approved
Encoding	Unicode (UTF-8)
Author	lokesh v [Change]
Description	
Created By	lokesh v, 16/11/2024, 12:09 pm
Modified By	lokesh v, 16/11/2024, 12:09 pm

The "Email Template" section shows the subject "leave approved" and the plain text preview:

dear({Tenant__c.name}).

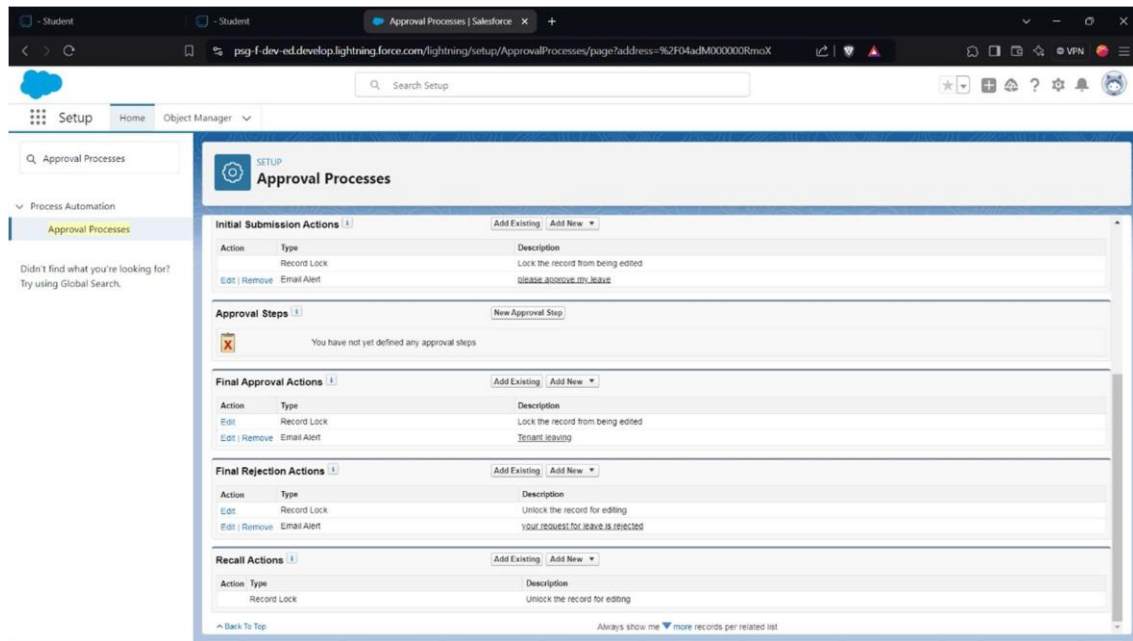
I hope this message finds you well . i am writing to inform you that I have received your email conforming the approval of my leave request. I would like to express my gratitude for considering and approving my time off

Your leave is approved . you can leave now



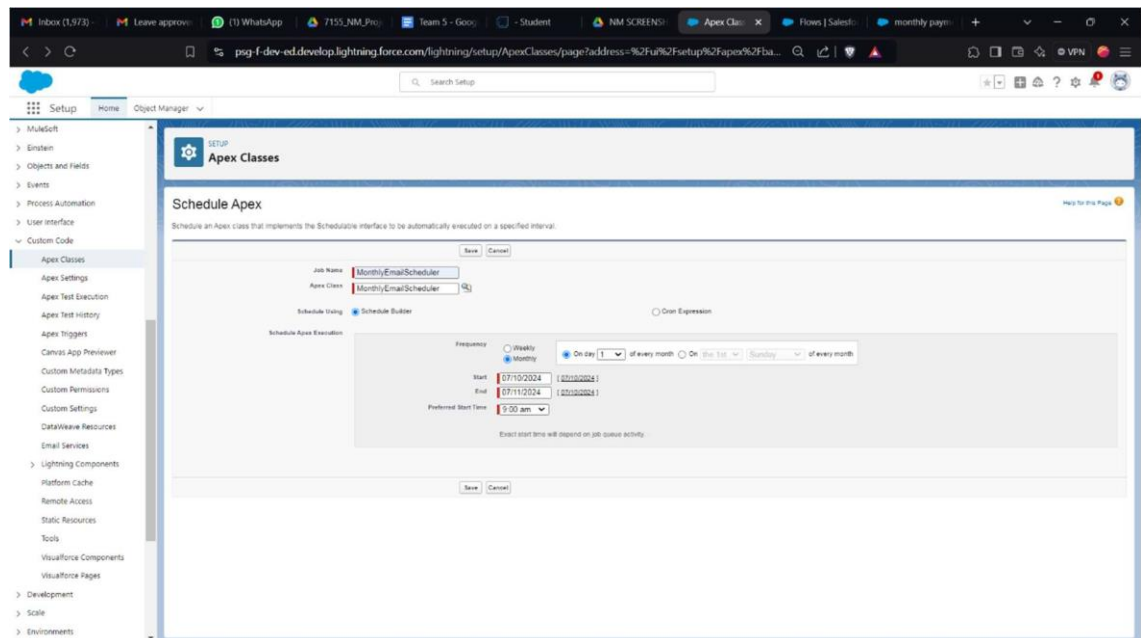
9. Approval Process for Vacant Properties

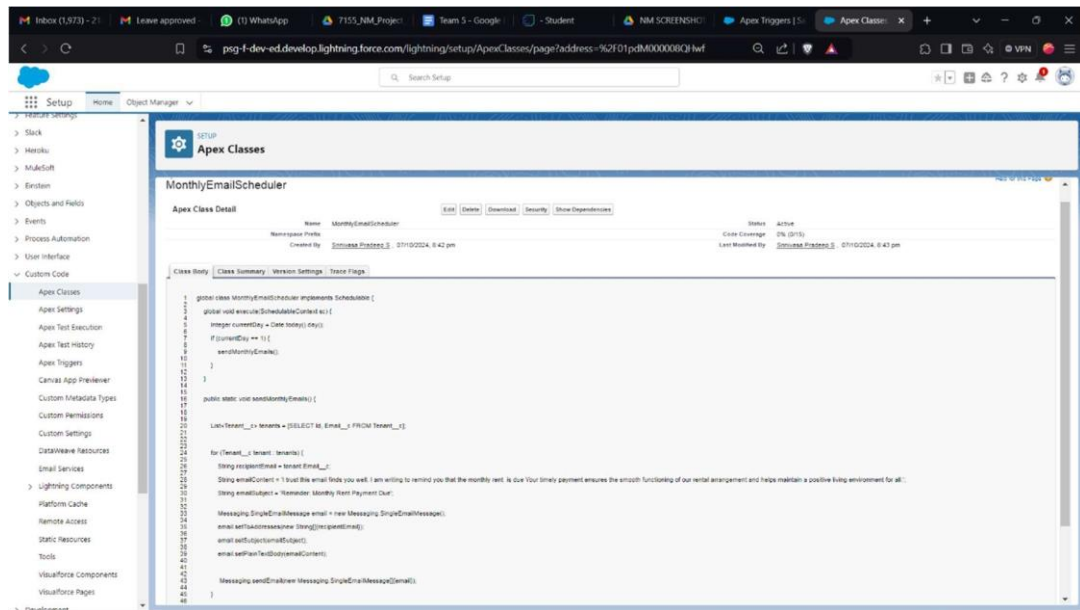
- Established an approval process to verify property availability before leasing:
 - Configured initial submission, final approval, and rejection actions to streamline the lease request process.



10. Scheduled Apex Class for Monthly Reminders

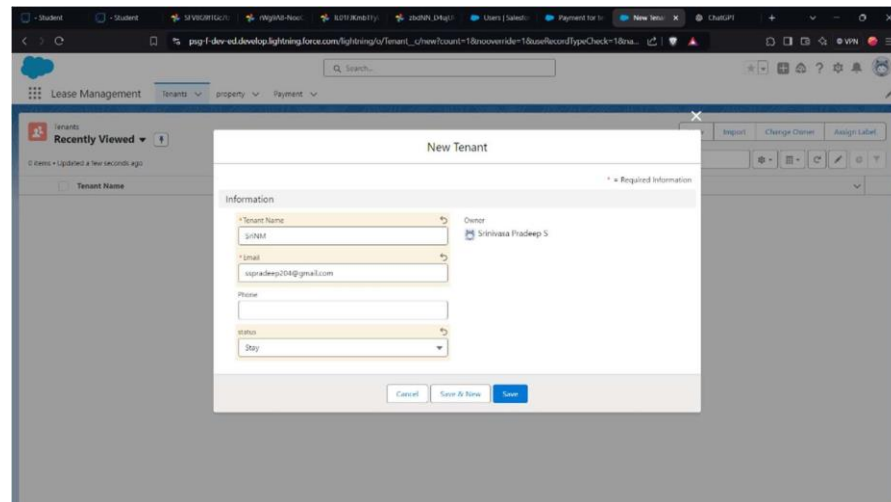
- Scheduled an Apex class to handle monthly billing reminders and lease renewals, ensuring tenants and administrators are up-to-date on upcoming obligations.





11. Create a Tenant Record

- Created a **Tenant** record within the system, entering all relevant details such as contact information, lease start and end dates, and tenant status.
- Ensured that the new tenant record is associated with the appropriate **Property** and **Lease** records through lookup relationships, enabling seamless data access and association.



12. Submit for Approval

- Initiated the **Approval Process** for the new lease by submitting the **Lease** record for approval.
- The submission triggers an initial review, allowing the leasing team to validate all provided information before final approval.

Lease Management

Tenant: Lokesh V

Related Details

Tenant Name: Srinivas

Email: Lokeshv204@gmail.com

Phone:

Status: Stay

Created By: Lokesh v. S. 07/10/2024, 9:23pm

Owner: Lokesh v.

Last Modified By: Lokesh v. S. 07/10/2024, 9:23pm

Activity

Filters: All time • All act

Refresh

Upcoming & Overdue

No activities to show.

Get started by sending an email, scheduling a meeting, or creating a task.

No past activity. Past meetings and tasks marked as completed.

Submit for Approval

13. Wait for Approval

- The **Approval Process** waits for the approver's action (approval or rejection). During this phase, the lease status remains in a "Pending Approval" state until final action is taken.

Lease Management

Tenant: Lokesh

Related Details

Payments (0)

Approval History (2)

Step Name	Date	Status	Assigned To
check for vacant	07/10/2024, 9:37 pm	Pending	Lokesh v
Approval Request Submitted	07/10/2024, 9:37 pm	Submitted	Lokesh V

View All

Payment (0)

Notifications

Lokesh V is requesting approval for tenant

Tenant Name: srinivas • Owner: Srinivasa Pradeep S

a minute ago

Jose Figueroa is requesting approval for research proposal

Research Proposal: Sole Society: Greener Kicks • Owner: María Concepción Morales • Created By: María Concepción Morales

20-Sept-2024, 6:14 pm

New Guidance Center learning resource available

Automate Processes with Workflow Tools

Get to know Salesforce's automation tools and discover which workflows will best help you reach your business goals.

18-Sept-2024, 12:13 pm

New Guidance Center learning resource available

Get to Know Your Customers

Understanding how accounts and contacts work together in Salesforce is essential to know who your customers are, where to find them, how to contact them, and how you can

The screenshot displays the Salesforce Lease Management interface. The top navigation bar includes 'Lease Management', 'Tenants', 'property', and 'Payment'. The main content area shows the tenant 'Lokesh' with tabs for 'Related' and 'Details'. Under 'Related', there are sections for 'Payments (0)' and 'Approval History (2)'. The 'Approval History' table lists two steps: 'check for vacant' (Approved, 07/10/2024, 9:38 pm) and 'Approval Request Submitted' (Submitted, 07/10/2024, 9:37 pm). A notification panel on the right shows a message: 'Approval request for the tenant is approved' by 'srinivas' a few seconds ago. Below this, there are two notifications about 'New Guidance Center learning resource available'.

14. Email Sent for Approval Confirmation

- Upon approval, an automated **Email Template** for lease approval is triggered and sent to the tenant, confirming the lease's approval status.
- The email outlines the lease's terms and conditions, ensuring clear communication with the tenant.

The screenshot shows a Gmail inbox with an email titled 'request for approve the leave' from 'Lokesh V via upamfits969m.dm-cjeavualind136.bnc.salesforce.com' to 'me'. The email content includes 'Dear Srinivasa Pradeep S,' and 'Please approve my leave'. The email is dated '21:37 (17 minutes ago)'.

The screenshot shows a Gmail inbox with an email titled 'Leave approved' from 'Lokesh V via iowkggljdpb.dm-cjeavual ind136.bnc.salesforce.com' to 'me'. The email content includes 'I hope this message finds you well. I am writing to inform you that I have received your email confirming the approval of my leave request. I would like to express my gratitude for considering and approving my time off. your leave is approved. You can leave now.' The email is dated '21:38 (16 minutes ago)'.

5. Testing and Validation

Unit Testing (Apex Classes, Triggers)

Apex Trigger:

trigger test on Tenant__c (before insert)

```
{
    if(trigger.isInsert && trigger.isBefore){
        testHandler.preventInsert(trigger.new);
    }
}
```

Test Class:

```
public class testHandler {
    public static void preventInsert(List<Tenant__c> newList) {
        Set<Id> existingPropertyIds = new Set<Id>();
        for (Tenant__c existingTenant : [SELECT Id, Property_c FROM Tenant__c WHERE
Property_c != null]) {
            existingPropertyIds.add(existingTenant.Property__c);
        }
        for (Tenant__c newTenant : newList) {
            if (newTenant.Property_c != null &&
existingPropertyIds.contains(newTenant.Property_c)) {
                newTenant.addError('A tenant can have only one property');
            }
        }
    }
}
```

MonthlyEmailScheduler:

```
global class MonthlyEmailScheduler implements Schedulable {

    global void execute(SchedulableContext sc) {

        Integer currentDay = Date.today().day();

        if (currentDay == 1) {

            sendMonthlyEmails();

        }

    }
}
```

```

public static void sendMonthlyEmails() {

    List<Tenant__c> tenants = [SELECT Id, Email__c FROM Tenant__c];

    for (Tenant__c tenant : tenants) {

        String recipientEmail = tenant.Email__c;

        String emailContent = 'I trust this email finds you well. I am writing to remind you that the monthly rent is due. Your timely payment ensures the smooth functioning of our rental arrangement and helps maintain a positive living environment for all.';

        String emailSubject = 'Reminder: Monthly Rent Payment Due';

        Messaging.SingleEmailMessage email = new Messaging.SingleEmailMessage();

        email.setToAddresses(new String[]{recipientEmail});

        email.setSubject(emailSubject);

        email.setPlainTextBody(emailContent);
        ;

        Messaging.sendEmail(new Messaging.SingleEmailMessage[]{email});

    }

}
}
}

```

Steps:

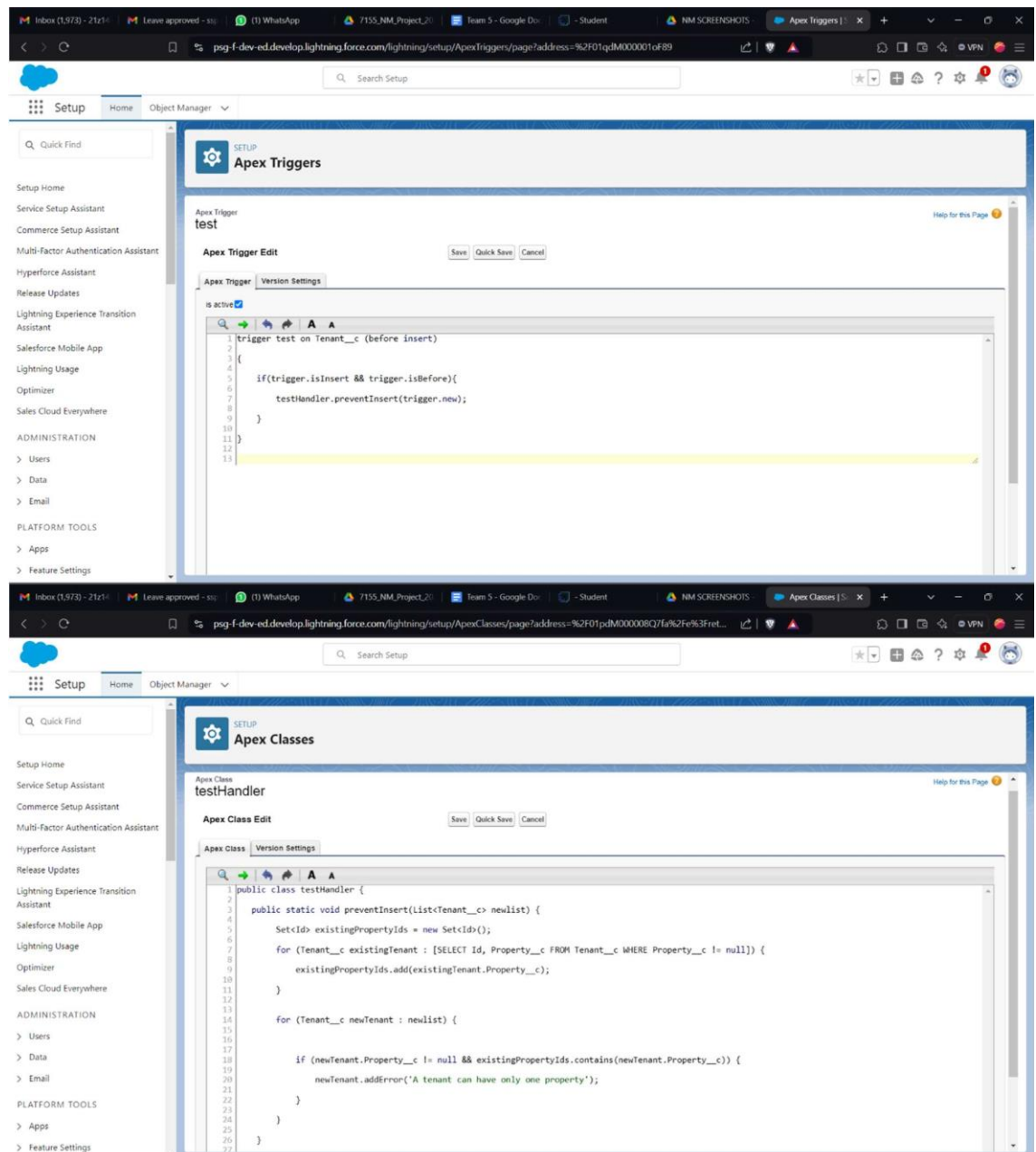
1. Created a Test Property Record: Created a Property__c record to serve as the property for assigning tenants.
2. Inserted the First Tenant: Inserted a Tenant__c record linked to the test property, simulating the initial tenant assignment.
3. Prepared a Second Tenant for the Same Property: Created another Tenant__c record with the same property ID to test the trigger's restriction on multiple tenants per property.
4. Inserted the Second Tenant: Attempted to insert the second tenant within a try-catch block to catch the expected error, confirming that the trigger prevented the insertion.

5. Verified Error Message: Checked if the error message matched the expected output: "A tenant can have only one property," verifying that the trigger works as intended.

6. Run Apex Test Execution:

- Go to Setup > Apex Test Execution.
- Select TenantTriggerTest and click Run.

Confirm that all assertions pass, indicating that the trigger prevents multiple tenants from being assigned to the same property.



Logs							
User	Application	Operation	Time	Status	Read	Size	
Srinivasa Pradeep S	Unknown	ApexTestHandler	10/7/2024, 11:56:46 PM	Success	Unread	5.49 KB	
Srinivasa Pradeep S	Unknown	ApexTestHandler	10/7/2024, 11:56:43 PM	Success	Unread	485 bytes	
Srinivasa Pradeep S	Unknown	ApexTestHandler	10/7/2024, 11:56:43 PM	Success	Unread	2.18 KB	
Srinivasa Pradeep S	Unknown	ApexTestHandler	10/7/2024, 11:56:43 PM	Success	Unread	18.06 KB	
Srinivasa Pradeep S	Unknown	ApexTestHandler	10/7/2024, 11:56:43 PM	Success	Unread	1.48 KB	
Srinivasa Pradeep S	Unknown	ApexTestHandler	10/7/2024, 11:56:43 PM	Success	Unread	23.35 KB	

6. Conclusion

Summary of Achievements: The Lease Management project successfully established an efficient, transparent, and scalable system for managing property leases. Key accomplishments include:

- **Organized Data Management:** Created custom objects, fields, and tabs for structured lease tracking.
- **Automation of Lease Processes:** Implemented triggers, validation rules, and scheduled classes to streamline recurring processes like monthly payments and notifications.
- **Improved Communication:** Developed approval processes and email templates for automated tenant communication, improving clarity and tenant experience.
- **Enhanced Decision-Making:** Built custom reports and dashboards to provide stakeholders with real-time insights, facilitating faster, more informed decisions.

This project illustrates the effective use of Salesforce in supporting lease management processes, driving operational efficiency, and ensuring data accuracy and compliance. Through this system, the lease management workflow has been optimized for better tracking and tenant relations.